

A Review of Catalyst Integration in Hydrothermal Gasification

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Abstract: Industrial scale-up of hydrothermal supercritical water gasification process requires catalytic integration to reduce the high operational temperatures and pressures to enhance controlled chemical reaction pathways, product yields, and overall process economics. There is greater literature disparity in consensus on what is the best catalyst and reactor design for hydrothermal gasification. This arises from the limited research on catalysis in continuous flow hydrothermal systems and rudimentary lab-scale experimentation on simple biomasses. This review summarizes the literature status of catalytic hydrothermal processing, especially for continuous gasification and in situ catalyst handling. The rationale for using low and high temperatures during catalytic hydrothermal processing is highlighted. The role of homogeneous and heterogeneous catalysts in hydrothermal gasification is presented. In addition, the rationale behind certain designs and component selection for catalytic investigations in continuous hydrothermal conversion is highlighted. Furthermore, the effect of different classes of catalysts on the reactor and reactions are elaborated. Overall, design and infrastructural challenges such as plugging, corrosion, agglomeration of the catalysts, catalyst metal leaching, and practical assessment of catalyst integration towards enhancement of process economics still present open questions. Therefore, strategies for catalytic configuration in continuous hydrothermal process must be evaluated on a system-by-system basis depending on the feedstock and experimental goals.

Keywords: hydrothermal technology; catalysis; feedstocks; continuous operation



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1. Introduction

Conventionally, gasification of different feedstocks to syngas (i.e., H₂, methane, and carbon oxides) takes place by a thermal process of partial oxidation or pyrolysis. These thermal techniques utilise dry feedstock of moisture content less than 10 wt.% [1]. Many challenging feedstocks such as municipal solids waste, sludge, and other organic wastes contain elevated amount of moisture and thus need unique technology for processing. Hydrothermal gasification (HTG) solves the challenge of high-water-containing feeds to achieve the same syngas compositions, and this improves the thermal technology efficiency and commercial viability for future scale-up [2]. Among synthesis gas (syngas) is hydrogen, which is mainly produced by steam reforming of naphtha and natural gas [3]. However, environmental degradation associated with utilization of fossil fuels has propelled hydrogen production from other feedstocks. Hydrogen production from biomass using different thermal and biological techniques has been investigated for a long time [4–6]. However, the challenge of high water content in the feed that requires extra energy during the drying process, among other factors, has limited the commercialization of these technologies. Hydrothermal conversion technology utilizes subcritical and supercritical water environments to covert high-moisture feeds including slurries and other organics such as petroleum-based waste to valuable products such as syngas [7,8].

Hydrothermal gasification can be interpreted as a catalytic process due to the unique properties of the reaction media. Water acts as a reaction media and at supercritical conditions behaves as a catalyst and reactant, thanks to its H⁺ and [−]OH that are capable of hydrolyzing many compounds without further catalyst addition [9]. This is beneficial for

mass transfer and allows for faster and more even mixing of reactants to obtain the desired reaction site due to the low viscosity and fewer hydrogen bonds in supercritical water [10]. Research in the hydrothermal conversion area has identified that utilization of catalysts is highly beneficial for hydrothermal gasification to hydrogen [11–13]. The review focuses on catalysts and their integration in hydrothermal conversion technologies while focusing on hydrothermal gasification.

2. Role of Temperature and Catalysts in Hydrothermal Conversion Systems

The operational temperature regimes are an important factor in catalytic hydrothermal gasification of different feedstocks. There are three temperature regimes identified in literature for the hydrothermal gasification process, and different catalysts are suitable for each.

2.1. Supercritical High Temperature—Regime I

This regime is defined as between 500–700 °C. Temperatures above 700 °C are possible but unlikely due to material limitations in the metal alloys typically used for hydrothermal reactors. This regime is dominated by the second-order reaction and the free radical mechanisms such as pyrolysis that are largely prevalent in low-density fluids. Activated carbon and alkali catalysts are mainly used in this temperature regime. Xu and their team investigated catalytic gasification of glucose in a continuous setup between 600–700 °C and at pressures between 25–30 MPa [14]. The activated carbon catalysts in the temperature regime were effective in inhibiting char formation while producing H₂, CH₄, CO, and CO₂. In addition, at these elevated temperatures, the amount of CO produced was proportional to the activated carbon catalyst used. Another hydrothermal gasification study at 700 °C and 28 MPa in a flow reactor yielded H₂, CO₂, and methane [15]. Hydrogen yield increased exponentially from actual reaction temperatures of 600 °C in a 1 L Inconel-625 continuous reactor to simulated temperatures up to 900 °C [16]. Furthermore, hydrogen presumably acted as a promoter for the decomposition of organics via hydrogenation. Schmieder and their team investigated natural fiber model compounds at 550–600 °C in continuous reactor, yielding mainly hydrogen and carbon dioxide [17]. In the early 2000s, hydrothermal gasification at higher temperatures gained momentum, especially with plant-based waste feedstocks; however, since then, limited research has progressed, plausibly due to reactor material failures, safety concerns, and economic viability at such high temperatures. Lately, a few studies on high-temperature hydrothermal gasification have been reported in the literature. For instance, Greta et al. investigated co-hydrothermal gasification of wet algae and algae hydrochar in a plug flow reactor at 650 °C and 30 MPa [18]. In addition, others studies at 625 °C in a batch reactor have been reported for hydrothermal gasification of waste tires using transition metal catalysts [19].

2.2. Supercritical Intermediate Temperature—Regime II

Regime II hydrothermal gasification occurs just above the critical saturation line of water in the range of 374–500 °C. The ionic reaction mechanism dominates most reactions away from the free-radical reactions because of the decrease in water density. This temperature range has been studied extensively for carbohydrates and lignocellulosic hydrothermal gasification; thus, the decomposition of cellulose and lignin have been understood to undergo hydrolysis with aid of metal catalysts in this temperature zone [20,21]. Most studies demonstrated batch reactor operations for this temperature range. Youjun et al. investigated hydrothermal gasification of glucose at 400 °C in an autoclave, and their results showed hydrogen as the main product enhanced by application of a nickel-based catalyst [22]. Onwudili et al. studied catalytic hydrothermal gasification of microalgae in a batch Inconel reactor at 500 °C and 36 MPa, aiming to produce enhanced yield of hydrogen [23]. In addition, Dreher et al., in 2012, designed a continuous flow reactor for material absorption or diffraction in supercritical fluids at 400 °C and 30 MPa [24]. Adar and their team investigated a combination of regime I and II temperatures (450–650 °C)

for supercritical water gasification of sewage sludge in a continuous plug flow reactor [25]. In addition, Chen et al. investigated hydrogen production from sewage sludge between 480–540 °C and 25 MPa in a fluidized bed reactor [26].

2.3. Subcritical below 374 °C of Water—Regime III

There is no phase distinction in this temperature zone: both liquid and vapor phases can co-exist. The equilibrium composition of gases is dependent on the feedstock, and some studies have reported mainly carbon dioxide and methane for many biomasses [27]. Gasification of high molecular compounds within this regime is challenging owing to slow reaction rates compared to supercritical temperature zones [28]. Gasification in the lower temperature regimes is critical for the technology's commercial viability since it requires low thermal energy input. The challenge remains on usage of catalysts to improve the system performance given the would-be slow gasification rates. Hydrothermal gasification in this temperature regime was investigated in a continuous-flow plug reactor at 348 °C and 20 MPa to yield hydrogen and other syngas [29]. Elliot et al. studied this temperature regime for over a decade since 1993 for temperatures near 350 °C and 21 MPa using different catalysts on different feedstocks such as manure and lignocellulosic waste on a bench flow reactor [30–33]. Nanda et al. studied the hydrothermal gasification of pinecones in subcritical conditions (300–370 °C, 21 MPa) for the production of hydrogen [34]. Seif et al. studied hydrogen production from distillery waste between 300–375 °C in a batch system, and the hydrogen concentration reached 48.81 mol% at 375 °C [35]. Subcritical hydrothermal gasification is often investigated with aid of catalysts and is still limited to non-complex feedstocks and model compounds. For instance, Minowa et al. investigated the hydrothermal gasification of cellulose between 200–350 °C in an autoclave with aid of alkali and nickel catalyst, and the highest H₂, of about 80 mmol, was produced at 350 °C [36–39].

3. Catalysts in Hydrothermal Gasification Systems

Catalysts in hydrothermal gasification play a crucial role in enhancing gasification efficiencies, moderate reaction conditions, gas selectivity, and yields. Catalysts do so by regulating specific reaction mechanisms during the reaction [40,41]. However, application of catalysts can have some disadvantages such as deactivation, plugging, fouling, and accelerated corrosion of equipment at elevated temperatures. Different classes of catalysts have been deployed for investigation of various hydrothermal gasification conditions: homogeneous catalysts and heterogeneous catalysts. It is worth noting that catalytic hydrothermal gasification can achieve 5–10% H₂ and 60–70% methane whereas high-temperature hydrothermal gasification has the potential to yield 20–50% H₂, 30–40% methane, and almost 12% useful hydrocarbons [42].

3.1. Homogeneous Catalysts

Homogeneous catalysts exist in the same phase as the reacting substrate. In hydrothermal supercritical gasification, metal alkali and oxides are generally used on different feedstocks. Homogeneous catalysts have advantages of being cheap and avoid issues such as plugging and sintering; however, they pose a disadvantage of poor recovery and hence have limited application in continuous reactors on the commercial scale [43]. This type of catalysis is known to facilitate the cleavage of carbon–carbon bonds in the substrate through a water–gas shift reaction involving water and carbon monoxide, hence increasing the hydrogen yield [44]. The presence of a catalyst can further increase gas yields by promoting the reaction of hydrogen with the intermediate products [45].

Supercritical water gasification investigations were performed at lab-scale using glucose as the feedstock in a continuous-flow reactor with homogeneous catalysis involving 0.1 wt.% KOH catalyst with feed concentrations between 0.25–2 wt.% at supercritical conditions of 450–560 °C, 25 MPa. The presence of the catalyst enhanced hydrogen production and H₂ gasification efficiency to 112.74% [45]. Rönnlund et al. investigated alkali catalysts

such as KOH, NaOH, and K_2CO_3 for supercritical water gasification of black liquor and paper sludge in a continuous reactor at 25 MPa and 500–650 °C. Hydrogen yield increased with increased alkali concentrations. Furthermore, alkali selectivity was key with waste type, where K_2CO_3 and NaOH produced more H_2 for paper sludge and black liquor, respectively [46]. Feng et al. studied a mixture of alkali (K_2CO_3 , KOH, Na_2CO_3 , NaOH) and activated carbon catalysts for supercritical gasification of sewage sludge in a batch reactor at 450 °C and 23–26 MPa. Potassium-based catalysts were effective for increasing gas yield and desulfurization of the syngas [47]. In addition, KOH and K_2CO_3 were superior to NaOH and Na_2CO_3 in producing more hydrogen during the gasification of perennial timothy grass in a batch reactor at 650 °C and 23–25 MPa. Notably, NaOH was found to favour methanation to form CH_4 at the expense of H_2 [48]. Sinağ and their team found K_2CO_3 to have better catalytic performance than Raney nickel catalyst during the gasification of 5 wt.% glucose in a batch reactor at 500 °C and 30 MPa [49]. Potassium-based catalyst has been proven to be better for enhancing hydrogen production for homogeneous catalysis on various feeds for hydrothermal gasification reactions [50,51]. Other important studies highlighting the mechanisms of reaction involving homogeneous catalysts are summarized in Table 1.

Table 1. Summary of other notable studies involving homogeneous catalysts.

Catalyst	Conditions	Feedstock	Comments	Reference
NaOH, Na_2CO_3 Ca(OH) ₂	• 400 °C • 25 MPa • 20 min • 1 wt.% catalyst • 6 wt.% feed • Batch	Sedum plumbizincicola	• NaOH had the best H_2 yield enhancement. • NaOH promoted CO_2 capture. • Catalyst enabled heavy metal stabilization.	[52]
KOH, K_2CO_3 , Na_2CO_3 , Ca(OH) ₂ , NaOH	• 700 °C • 25 MPa • 5 min • 2–20 wt.% catalyst • 2–20 wt.% feed • Batch autoclave	Coal (100–150 µm)	• Hydrogen increased in order of $K_2CO_3 > KOH > NaOH > Na_2CO_3 > Ca(OH)_2$	[53]
Red-mud, Ranney Ni, trona, K_2CO_3	• 500 °C • >50 MPa • 60 min • 0.8 g catalyst • 8.3 g feed • Batch autoclave	Cotton stalk, corncob, tannery waste (<200 µm)	• Catalyst effect varied with the type of biomass. • Increased hydrogen with catalyst addition. • Trona and K_2CO_3 had similar gasification activity.	[54]
KOH, NaOH, Na_2CO_3	• 500–700 °C • 22.5 MPa • 30 min • 0.1–0.3 wt.% catalyst • 0.5–2.5 wt.% feed • Continuous Inconel 625 reactor	Lactose monohydrate	• Catalyst significantly increased the H_2 yields and decreased production of CO. • Catalysts significantly reduced sample total organic carbon in order of $Na_2CO_3 > KOH > NaOH$.	[55]
KOH, K_2CO_3 , H_2O_2	• 400 °C • 22 MPa • 1–30 min • 38 g feed • Batch	Coking sludge	• KOH enhanced hydrogen production.	[56]

3.2. Heterogeneous Catalysts

Heterogeneous catalysts are widely used on small, complex, and commercial processes. This category of catalysis enjoys several advantages over homogeneous catalysts such as the fact that they are easy to recover after reaction, provide more active sites resulting in improved efficiency during gasification, and offer greater selectivity towards promotion of desired products from a specific reaction process [57,58]. Several studies in the available literature have been reported for heterogeneous catalysis with different feedstocks. Onwudili and Williams investigated supercritical water gasification of plastic waste while deploying mono- and bimetallic transitional metal catalysts such as gamma-supported Ni and Ru. Results from their batch system showed that a ruthenium-based bimetallic catalyst performed better than a nickel-based catalyst for hydrogen production. In addition, the order of decreasing gasification efficiency was as follows: polypropylene > high density polyethylene > low density polyethylene > polystyrene [59]. Karakuş et al. investigated platinum- and ruthenium-based catalysts for their catalytic performance during the gasification of 2-propanol in a continuous reactor. The platinum catalyst produced better selectivity for hydrogen than the Ru catalyst under the same investigated conditions [60]. Lu et al. investigated the performance of a range of catalysts such as MgO-supported Fe, Ni, Cu, and Ni with supports. Using wheat straw as substrate in a batch reactor, Ni > Fe > Cu for hydrogen production in MgO-supported metals. Furthermore, the results of nickel with other supports showed Ni/MgO to have the best H₂ yield [61]. Nickel- and ruthenium-based catalysts have been widely studied for their catalytic performance during hydrothermal gasification, mainly on glucose as the reactant in a batch system [41,62–64].

3.3. Transitional Metal Catalysis

Transitional metal catalysts (TMCs) are the most common metals used as catalysts for gasification in supercritical environments due to their superior performance towards the conversion of carbonaceous materials and the relatively low cost of the catalysts [65].

Nickel-based catalysts have widely been investigated for gasification of different feedstocks thanks to their high catalytic activity, gasification efficiency, and low temperature requirements during conversion studies. However, nickel was reported to increase H₂ consumption through the methanation reaction, yielding methane [66]. Farusawa et al. investigated the regenerative capability of nickel-based catalysts in a supercritical water gasification system. Nickel on supported MgO was reused three times before deactivation using lignin as the feedstock in a stainless steel bomb reactor [67]. Hydrogen selectivity was investigated over different supported TMCs: Ni, Co, Cu, and Ru on gamma alumina, zirconia oxide, and activated carbon supports; Ni and Ru produced the highest H₂ selectivity and catalytic activity [68].

Platinum-, molybdenum-, iridium-, rhodium-, palladium-, and zirconia-based catalysts have been investigated for HTG and their H₂ yield, and gasification efficiencies were lower than nickel- and ruthenium-based catalysts [69,70].

There are potential catalyst supports that have been proven to increase catalysts' stability and activity during HTG. For instance, AC, titanium dioxide, aluminium oxide, ceria, zirconium dioxide, etc. [69,71,72]. Other notable transition and noble metal catalysts used for HTG are listed in Table 2.

Table 2. Summary of notable transition metal catalysts for hydrothermal gasification.

Catalyst	Conditions	Feedstock	Comments	Reference
- Pt - Ir - Ni	450 °C 25 MPa 1.3 s residence time 1 g catalyst 5–30 wt.% feed - Continuous	Ethylene glycol	- Pt produced the best H₂. - Ni in Ni-Pt enhanced H₂ and reduced methanation. - Ni presence in a bimetallic composition enhanced catalyst stability.	[73]
Ru/ZrO₂	550 °C 35 MPa 1100 min 1.0% catalyst 5 wt.% feed - Continuous	Glycerol (<0.2 µm)	- Water-soluble products of acetaldehyde, acetic acid, hydroxyacetone, and acrolein. - H₂, CO main gases, CO₂, CH₄, and C₂H₄. - Catalyst recycling up to >18 h was investigated.	[74]
Ni and Ru on γ-Al₂O₃ and ZrO₂	500–700 °C 25 MPa 10–20 g/L feed 2 mL/min 1–5 h 0.04–10% catalyst load - Continuous	Cattle manure oil	- Ni10%-Ru0.08%/Al₂O₃-ZrO₂ 1.10 0.95 had H₂ yield (1.1 mol/mol_{carbonfeed}). - H₂, CO₂, and CH₄ were produced. - H₂ yield increased with time (1–5 h). - H₂ and CO₂ were highest at 700, 600, and 500 °C, respectively. CH₄ yield trend was in reverse order at the same temperatures. - H₂ and CO₂ yield increased inversely with feed concentration and the opposite for CH₄.	[75]
Raney Ni, K₂CO₃	650–800 °C 23–27 MPa 0.1–0.3 wt.% catalyst 3–7 kg/h feed rate - slurry 16 wt.% - Continuous Inconel 625	Coal	- H₂ yield double with Ni compared to K₂CO₃. - H₂, CO, CO₂, CH₄, and C₂ products. - Increase in temperature sharply increased gasification efficiency and H₂ yield. - Pressure has no significant effect on the gasification of coal in supercritical water.	[76]
Ni, Co, Ru on Ti, Zr-based supports	600 °C 10–25 MPa 2 h 0.6 mL/min 1.5 wt.% Ru, 10 wt.% Ni, Co, respectively - Continuous	Switch grass biocrude	- H₂, CO, CO₂, CH₄, and C₂H₆ were produced. - Ni/ZrO₂ and Ru/TiO₂ gave the highest H₂ yield of 0.98 and 0.82 mol/Mol C_{reacted}, respectively. Both catalysts had the lowest CH₄ yield. - Ni/TiO₂ had the lowest H₂ yield.	[71]
Ru/activated carbon	350–353 °C 20–21 MPa 6–10 h 1.5 L/h 7.8 wt.% Ru 17–35 wt.% dry solids - Continuous	- Algae slurry - HTL biocrude	- CH₄ > CO₂ > NH₃ > H₂ > C₂H₆ in higher vol% were produced. 5 g/L Ca (OH)₂ was added to the feed before entering the catalyst bed to precipitate sulphates that could poison the catalyst.	[77]

4. Effect of Catalysts on Reactor and the Gasification Reactions

Most catalysts improve H₂ yield through the favorable WGS reaction [78–80]. However, some catalysts have no effect on underlying gasification reactions such as dehydration, decarboxylation, and hydrolysis [20]. Moreover, little is known about the LCA (Life Cycle

Assessment) and TEA (Technoeconomic Analysis) for catalyst integration and its significance in improving the process economics or system longevity. Literature works on LCA and TEA focus on the hydrothermal biomass conversion technology [81]. Thus, there is a need to bridge the gap between catalyst integration and hydrothermal technology to overcome overall process economics hindering its commercial adoption. Different classes of catalysts present different challenges during catalytic hydrothermal gasification. For instance, alkali metals such as NaOH, Na_2CO_3 , KHCO_3 , etc., are effective catalysts; however, due to the salts' insolubility in supercritical water, increased corrosion, reactor fouling, clogging, and difficulty in recovery makes their use a challenge. Alkali salts interact with the reactor inner walls, and the mechanistic understanding of this phenomena is very limited. It is plausible that the salts dissolve the inner protective metal oxide layer of the reactor surface and the resultant corrosion products can act as catalysts during the reaction. Due to this assumption, alkali metal catalysts are not suitable for catalytic hydrothermal gasification studies [72].

Activated carbon (AC) is suitable for simple hydrocarbons like glucose and glycerol via water–gas shift reactions. However, this class of catalyst decomposes in supercritical water and deactivates within few hours of reaction and is thus not suitable for longer reaction time and hydrocarbons [14,82]. Rather than using AC as a gasification catalyst, it is advisable for use as a catalyst support for metal-based catalysts [83].

Sintering and deactivation over long-term use is the major challenge facing Ni-based catalysts. In addition, the catalytic activity of nickel is often compromised by carbon deposition during reaction. Lab-scale reactors with small-diameter tubing lead to increased surface-to-volume ratios (S/V) which creates a catalytic wall effect and contributes to reactor weakness [70,72]. Therefore, using higher-nickel-containing alloys or introducing dean vortices to optimise the S/V would accelerate nickel application in the HTG processes.

A ruthenium-based catalyst demonstrated high stability and ability to prevent char formation with minimal intermediate products over a range of feedstocks [84,85]. However, catalyst cost and catalyst poisoning from sulfur-containing feedstocks that limit the catalytic cleavage of the C–C bonds remain challenges [70,72].

It is important to note that during operation, the catalyst is often deactivated or depleted, and it is difficult to replenish without complete shutdown and disassembly of the reactor. The depleted catalyst can easily deposit metal compounds in the effluent, which are often toxic and must be dealt with through separation and/or neutralization [72].

5. Types of Reactors and Reactor Configurations Used for Catalytic Gasification

Most heterogeneous mono- and multi-metal-based supported catalysts are packed in the reactor as pellets with porous frits mounted in the reactor inlet or outlet [14,86]. There are notable continuous reactor designs with catalyst application towards HTG, as shown below.

Catalytic process application: The biomass, in the presence of a non-metallic catalyst, was pretreated at temperatures conditions of 100–250 °C and a pressure in the range of 0.1–4 MPa, respectively. The entire process layout is shown in Figure 1A. The adjustment unit and a cut-crush unit (Figure 1B(i)) were used to stir and mix the supplied biomass, water, and the non-metallic catalyst. Downstream of the adjustment unit was the crusher pump that was used for further size reduction and slurry delivery to the pre-treatment unit and the reactor. The rotating shaft serves as a rotor and a mechanical seal supported by the bearing portion. As the impeller rotates inside the casing, pressure-fed fluid emerges from the discharge nozzle attached to the suction passage side, which is the distal-end side of the rotating shaft, via a key. The crusher pump obtains particle size of 500 µm or less, more preferably the average particle size was 300 µm or less. The crusher is a cut and crush design system. The gasification temperatures and pressures exceeded 374 °C and 22 MPa, respectively. The system was designed to include a catalyst recovery unit (see Figure 1B(ii)).

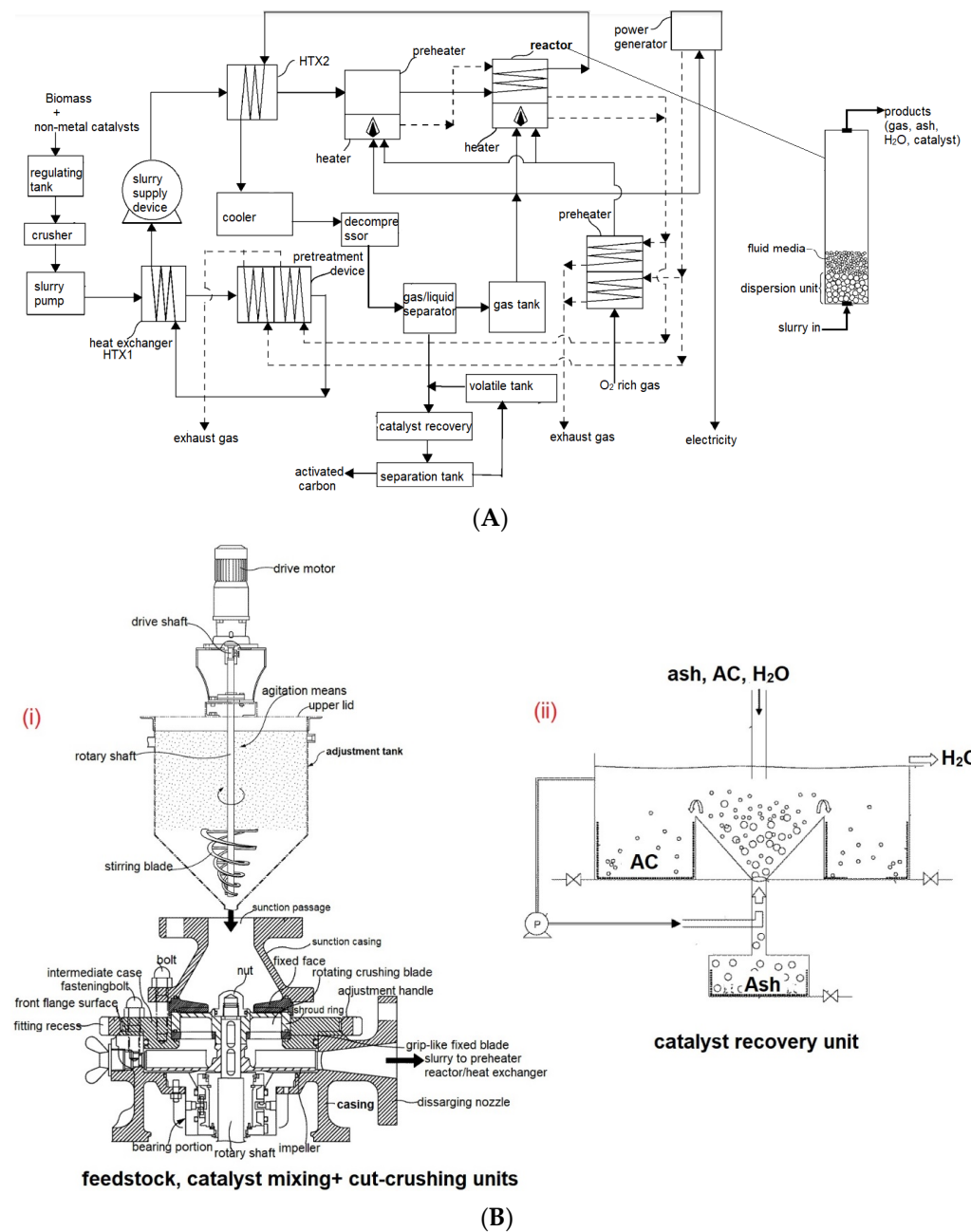


Figure 1. (A) Schematic layout of the patented supercritical water gasification of wet biomass [87]. (B) Schematic layout of selected unit processes in the patented catalytic gasification of wet biomass [87].

Catalyst application: As shown in Figure 2, catalyst application was disclosed in the invention relating to the biofuel-producing apparatus using a supercritical water gasification process. The high-pressure pump delivered the biomass to a tubular reactor that preheated it using an electric furnace and pressurized the biomass at 200–300 °C and 20–30 MPa. The supercritical water gasification reactor was operated at temperatures and pressures between 450 and 600 °C and between 20 and 30 MPa, respectively. The SCWG reactor was loaded with different metal catalysts such as zeolite, platinum (Pt), metal oxide–nickel composite catalyst, α -alumina catalyst, and platinum/ Al_2O_3 catalyst; in addition, the metal oxide–nickel composite catalyst was selected from the group consisting of chromium oxide (Cr_2O_3), alumina, and at least one metal oxide selected from the group consisting of iron oxide (Fe_2O_3) and nickel. The zeolite may include silica (SiO_2) and alumina (Al_2O_3). The

process included a gas scrubbing tank and a humidifier (humidifier made of activated carbon, silica gel, calcium chloride, zeolite, and sodium hydroxide that absorbed CO_2 and residual moisture in the gaseous products, respectively). The moisture absorber was filled with a non-precious metal catalyst. The Fischer–Tropsch Syn-gas to liquid reactor was packed with a non-noble metal catalyst of iron or cobalt and the reaction proceeded at conditions of 200–300 °C and 1 to 4 MPa to produce the desired liquid product. The reactor was coated inside with ceramics. C_1 – C_4 alkane, syn-gas, benzene, and toluene gas were produced.

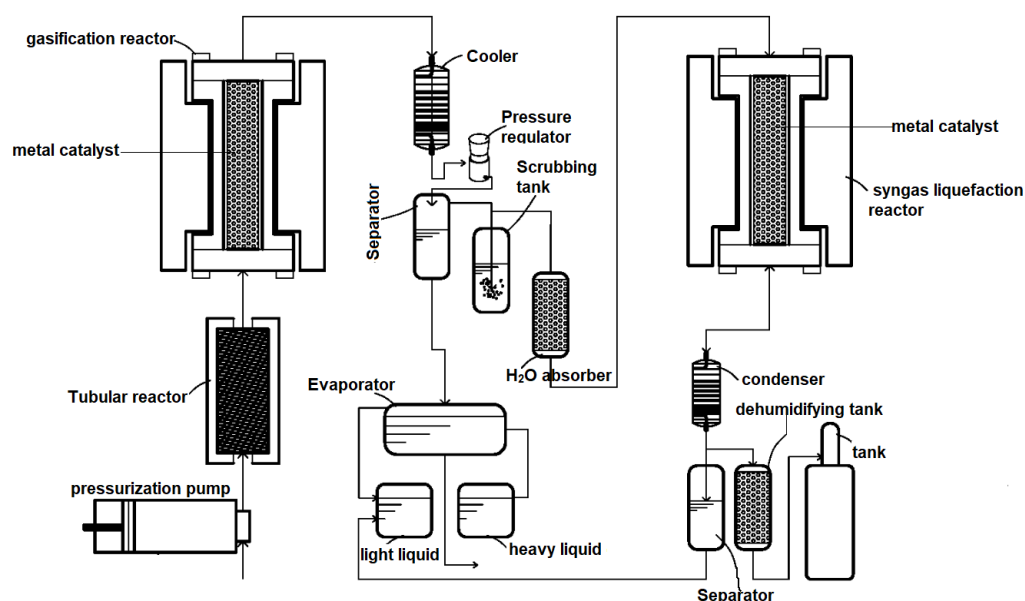


Figure 2. Schematic layout of the patented supercritical water gasification of algae biomass [88].

Catalyst application: In Figure 3, noble and base metal catalysts were studied in continuous supercritical conditions for catalytic reforming of ethylene glycol to hydrogen. Alumina-supported mono- and bimetallic iridium, platinum, and Ni catalysts were investigated for hydrothermal reactions at 450 °C and 25 MPa. As shown in Figure 3, the catalyst packed bed was used during catalyst reaction investigations. Before receiving the preheated feed, 1.0 g of catalyst was placed in a 63 cm long reactor (ID = 7 mm). Catalysts sizes ranged between 1.2 and 78 nm and loading between 0.1 and 1.45%. Platinum showed the highest hydrogen yields; however, catalyst stability and activity decreased with the decrease in the Pt loading. Also, the presence of Ni in a noble metallic configuration improved H_2 selectivity more than Ni alone.

Catalyst application: The tubular reactor in Figure 4 was 8.68 cm long and 0.635 cm in diameter (total volume 2.75 mL). A total of 1–3 g catalyst was packed with 3–5 g of inert particles (zirconia support) to improve heat transfer and decrease temperature gradients. The particles were supported onto a stainless-steel frit (10 m, VICI), and a thermocouple was inserted to record the temperature of the catalyst bed.

As shown in Figure 5A,B, 83.3 wt.% of slurry was fed in the reactor operated at 3 MPa, 1000 °C, 1–2 g per min, and 3 SLPM (atomization gas) was successfully investigated. For particles of 50–100 μm , no agglomeration was observed. The gas-assisted atomization of the slurry was achieved in a manner similar to liquid fuel injectors found in aerospace propulsion applications. The catalyst deployment mechanism was not explained outside the heat transfer and scalability principles.

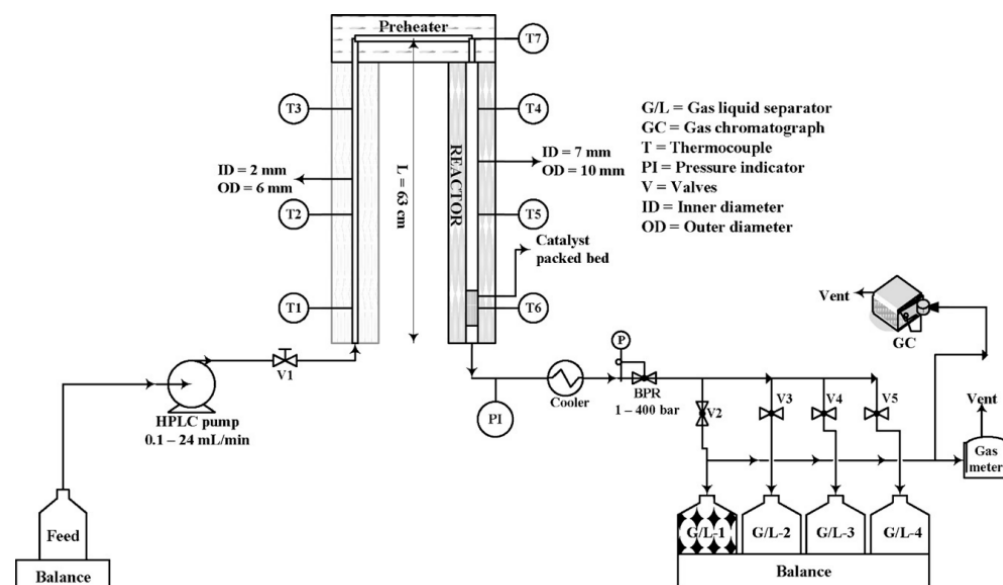


Figure 3. Continuous flow supercritical water reformer [73].

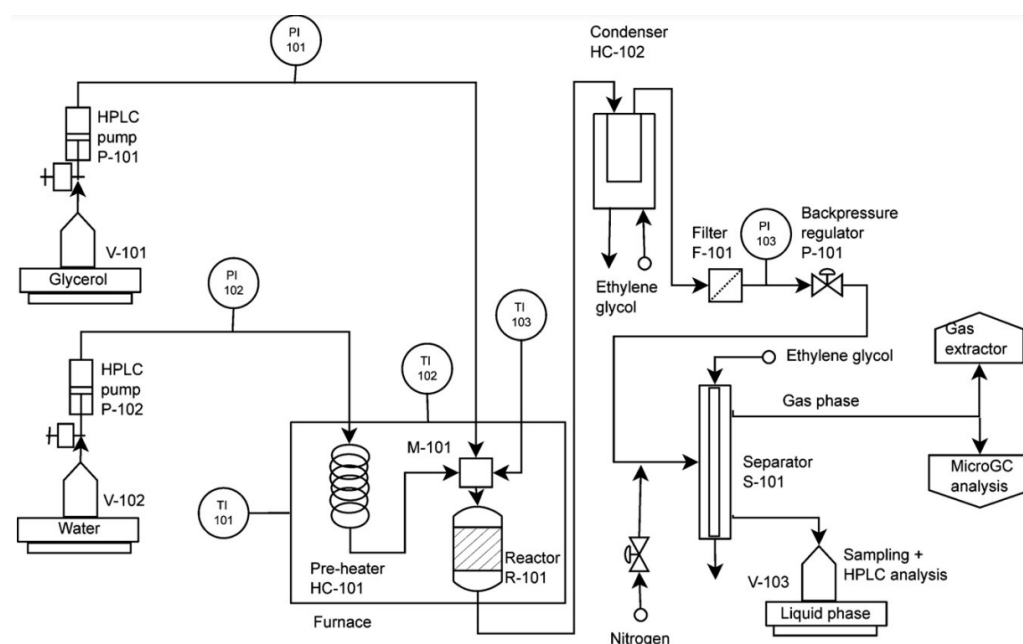


Figure 4. Continuous catalytic hydrothermal set of glycerol gasification [74].

Catalyst application and design: Inconel 625 tubular reactor (12.84 mm OD $\times$ 9.28 mm ID $\times$ 660.67 mm length) was packed with sieved catalyst supported on a “mesh 80” stainless steel membrane (see Figure 6). Quartz wool of 2 cm in depth was placed in between the catalyst support rod and the screen to avoid the possible carry-away of catalyst particles. Catalyst reduction was performed in-situ by supplying hydrogen at a rate of 5 mL/min for 90 min once the reactor reached the reaction conditions (500, 600, and 700 °C; 25 MPa). Using a HPLC pump, 10–20 g/L of HTL liquid from animal manure was fed into the system at 5 mL/min before lowering it to 2 mL/min to achieve the 2 h^{−1} space hourly velocity. Plugging challenges were cited as a widely experienced problem in continuous flow reactors due to tar and char deposition and agglomeration on the catalysts. Moreover, sintering (due to high temperature) of the particles, support material decomposition, and catalyst metal leaching are other frequent issues of using heterogeneous catalysts in continuous

flow reactors. Therefore, catalysts must not only focus on performance of the feed but also the intermediates.

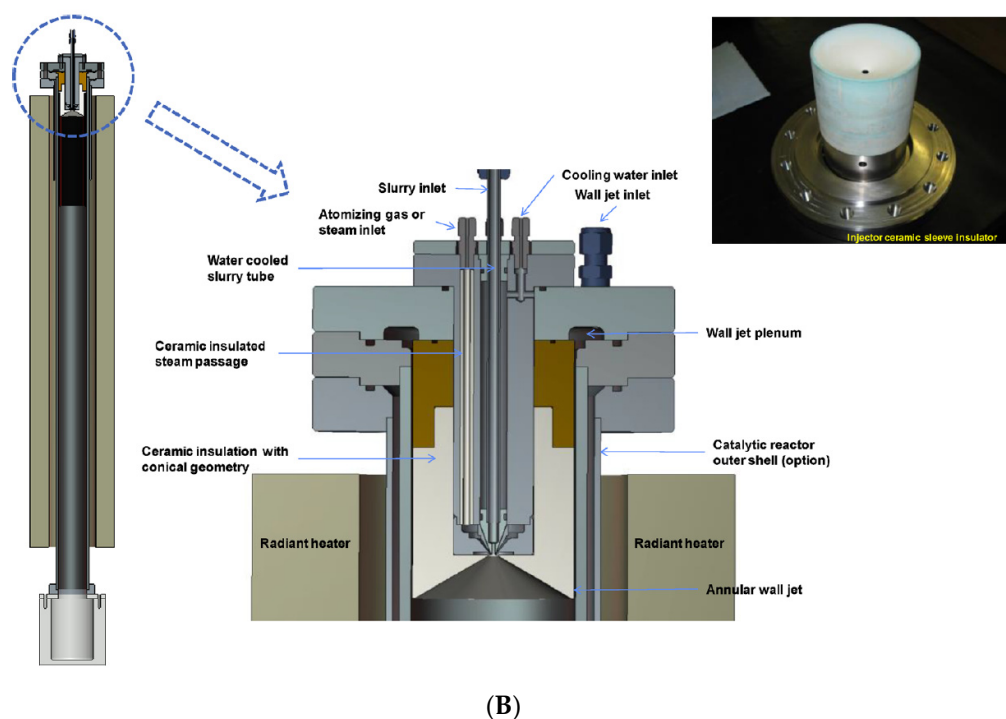
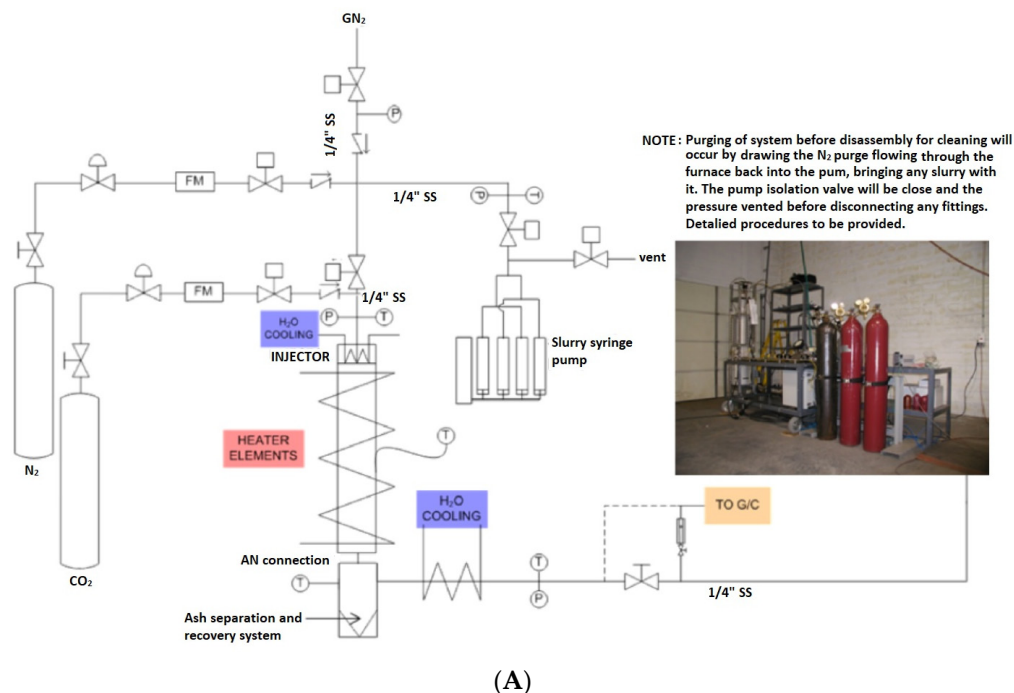


Figure 5. (A) Schematic layout of the experimental allothermal gasifier [89]. (B) Picture of the micronized coal slurry injector in the allothermal gasifier [89].

The reactor design was made of Hastelloy C276 tubing (17.15 mm OD, 10.85 mm ID and 1.24 m length) (see Figure 7). Flow rates of the coal slurry and pressurized water were measured by two mass flow meters (Endress + Hauser and Micro Motion). Raney nickel and K_2CO_3 catalysts were directly added to the slurry. Sodium carboxymethyl cellulose (CMC) was added as an additive to prevent precipitation during continuous delivery at high pressure. Operation conditions at 800 °C and >23 MPa were successfully investigated.

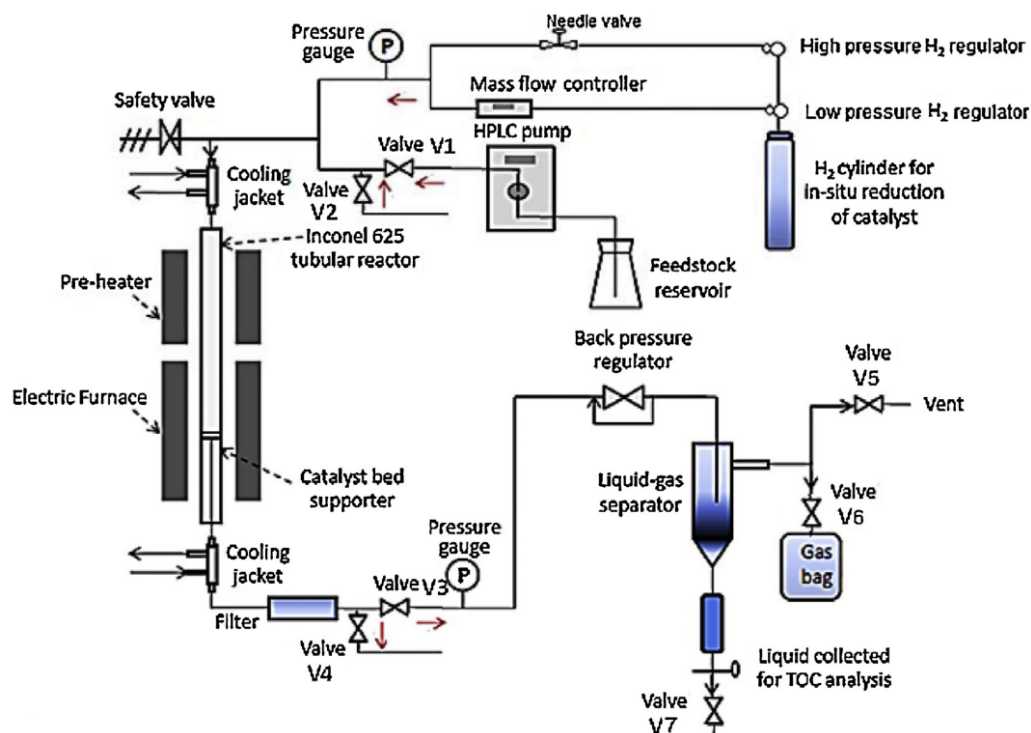


Figure 6. Schematic layout of SCWG reactor [75].

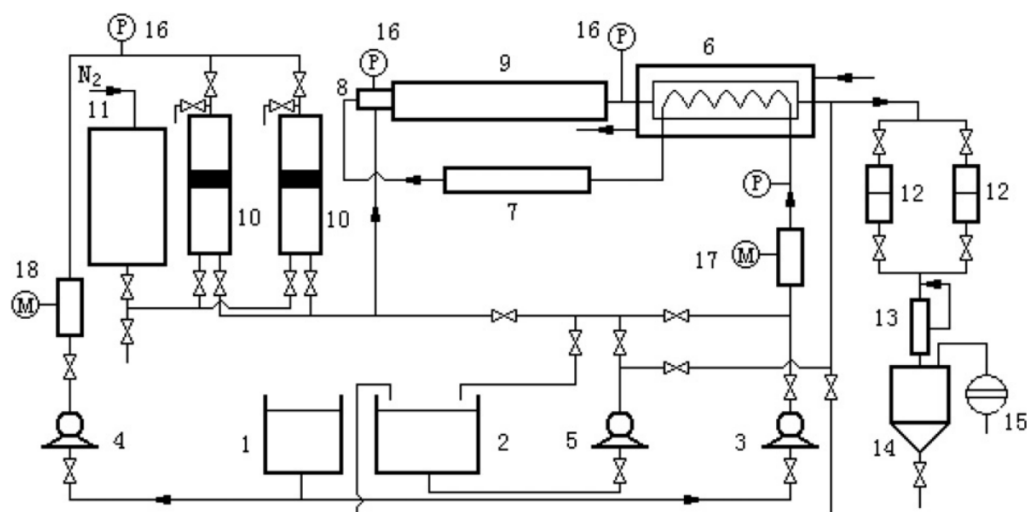


Figure 7. Schematic layout of SCWG of coal [76].

Catalyst application: In the Inconel 600 reactor (6.4 mm OD 3.2 mm ID 50 cm), 2 g of catalyst was packed into the reactor tube shown in Figure 8. The catalyst was retained in the reactor by placing stainless steel frits with a pore diameter of 0.5 μm (Valco) at either end. Following catalyst reduction, the reactor system was pressurized with water using two back pressure regulators in series reading 25 MPa and 10 MPa, respectively. The system temperature was raised to 600 $^{\circ}\text{C}$. After 1 h, the water in the reactor was replaced by the feed.

Catalyst application: The 412 mL fixed-bed catalytic hydrotreater in Figure 9 was made from 317 stainless steel (1" ID $\times$ 32" long). The bio-oil and hydrogen gas entered the top of the catalyst bed and passed downward through the bed, assumed to be in a trickle flow. The temperature of the catalyst bed was monitored by a thermocouple, which was adjustable to various points along the centerline thermowell. During catalytic hydrothermal gasification, the preheated feed from the CSTR passed through the solid

separator as well as a sulfur stripping bed before entering the up-flow, fixed catalyst bed in the 1 l tubular reactor. This combination of CSTR and plug-flow was used in these tests because of a conservative approach based on plugging problems experienced previously with a plug-flow-only reactor system with lignocellulosic feedstocks.

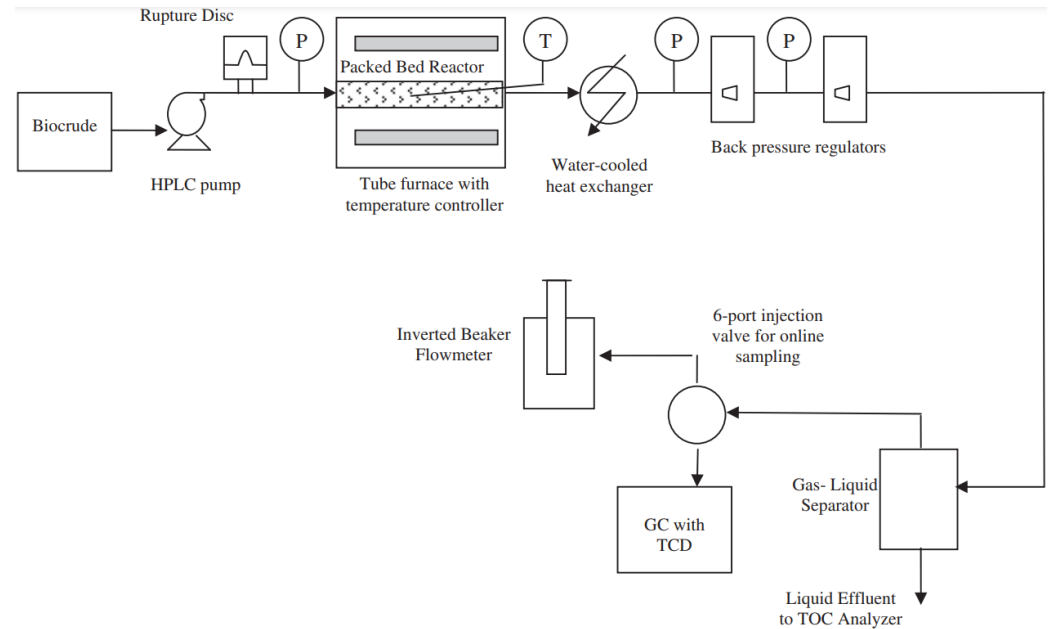


Figure 8. Schematic layout of SCWG of switchgrass biocrude [71].

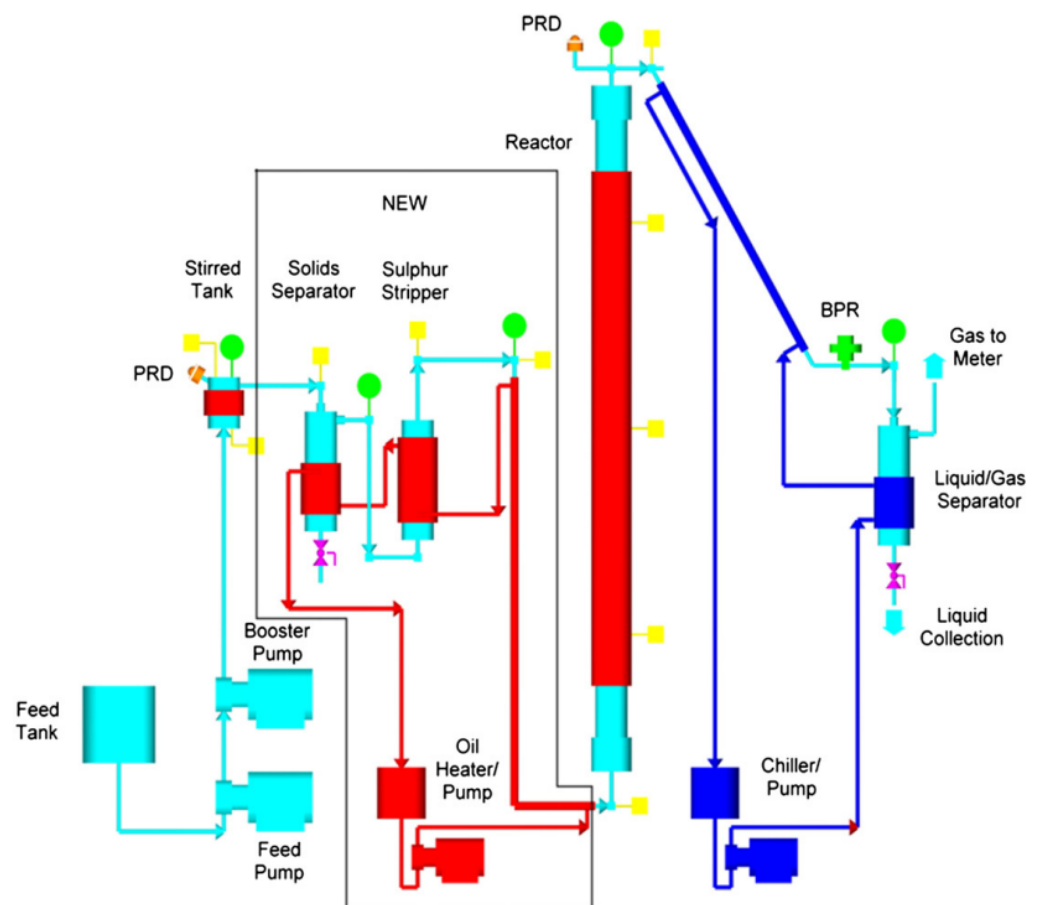


Figure 9. PFD of SCWG of switchgrass biocrude [77].

As shown in system schematics of literature Figure 10 [90], the process is comprised of feeding the wet biomass (sewage sludge, animal manures, fermentation residues, dredging sludge, algae) at a temperature of 760 °C and a pressure of up to 35 MPa to a reactor comprising a bubbling fluidised bed of solid particles suspended in a fluid, of which the bubbling fluidised bed has a length of 0.5 m and a cross-sectional area of at most 0.2 m². A worm pump or a lobe pump operating at low pressures were suggested for feeding the biomass to the first chamber. A high-pressure hydraulic pump such as a plunger pump or a membrane pump was then used to deliver the biomass to the second chamber.

6. Conclusions

The application of catalysts in continuous hydrothermal gasification enhances product yields; however, there are some challenges such as deactivation, plugging in continuous reactors, and fouling and accelerated corrosion of equipment at elevated temperatures.

Catalyst integration could theoretically enhance process economics; however, a detailed techno-economic assessment of catalyst use in SCWG is difficult due to the cost and complexity of catalyst preparation and the uncertainty of catalyst lifetime due to sintering and depletion. Designers must balance the material cost of a catalyst and integration costs against process improvements, such as lower reaction temperatures, higher viable solid loadings, and higher H₂ yields. In addition, the introduction of some catalysis may reduce corrosion by lowering operating temperatures; however, the process complexity is increased.

NaOH, Na₂CO₃, KHCO₃, and alkali metal salts are insoluble in supercritical water and can dissolve the inner protective metal oxide layer of the reactor surface, and the resultant corrosion products can act as catalysts; hence, they are not ideal for further catalytic HTG investigations.

Most heterogeneous mono- and multi-metal-based supported catalysts are packed in the reactor as pellets with porous frits mounted in the reactor inlet or outlet. Some reports highlighted the need for mixing inert particles such as zirconia or bids to ensure enhanced heat transfer and decrease heat gradients.

No catalyst was highlighted as the best choice for hydrothermal conversion. However, ruthenium-based and carbon-containing catalysts such as activated carbon catalysts were reported to have good activity and stability over a long reaction time in a supercritical water environment. It is important to note that during operation, the catalyst is often deactivated or depleted, and regeneration is difficult without complete shutdown and disassembly of the reactor. The depleted catalyst can easily deposit metal compounds in the effluent, which are often toxic and must be dealt with through separation and/or neutralization.

Most available literature investigations of catalytic hydrothermal gasification are limited to simple biomass and syringe pump, lab-scale systems with no investigations of end-of-life landfill waste including plastics.

Sintering and deactivation over long-term use is the major challenge facing promising Ni-based catalysts. In addition, the catalytic activity of nickel is often compromised by the carbon layer deposition during reaction. Many lab-scale, small-diameter tubular HTG reactors have high surface area-to-volume ratios (S/V), which creates an excessive catalytic wall effect that contributes to reactor weakness. Therefore, using higher-nickel-containing alloys or introducing dean vortices to optimise the S/V would accelerate nickel's application in the HTG processes.

The following recommendations have been suggested to counter some of the problems stated in the literature. (1) To counter corrosion and deter rapid erosion of the inner reactor wall, consideration should be given to using high-nickel stainless steels or nickel-based alloys or introducing dean vortices to optimise the S/V for catalytic HTG processes. Alloys such as Inconel 625 and Hastelloy C276 were reported to withstand operational temperatures of 800 °C and >23 MPa. Furthermore, inner coating of the stainless-steel reactor with a ceramic liner can help to preserve the protective nickel oxide layer and chromium and prevent it from influencing the catalytic reactions after long exposure.

(2) Metal-based catalysts containing transitional metals or zeolites with little doping on noble metals like ruthenium to create bimetallic catalysts capable of withstanding sintering and deactivation in the supercritical environment should be adopted. Catalysts must not only focus on conversion of the feed but also the intermediates and their ability to deter intermediates that are often the root-cause of operational difficulties. (3) Catalytic reactions should be investigated in a separate reactor, separated from particles that could lead to deactivation, either after the first reactor or following the second reactor in a two-reactor setup. This is likely to lower the operational challenges and system economics. (4) Special configurations such as a packed bed catalyst chamber, slits, and fluidization media might be needed for effective catalytic investigation of hydrothermal gasification of plastic-containing waste. (5) Alkali metal salts such as NaOH, Na₂CO₃, and KHCO₃ and activated carbon severely compromise the system functionality and require additional steps for recovery, and many neither offer regeneration or reuse; thus, they should not be considered for further investigations with a commercial scale-up mindset. (6) Life cycle assessment (LCA) and technoeconomic analysis (TEA) are needed to determine whether catalyst integration significantly improves the process economics or system longevity.

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Nomenclature

HTG	Hydrothermal gasification
HTL	Hydrothermal liquefaction
SCWG	Supercritical water gasification
Syngas	Synthesis gas
H ₂	Hydrogen
CH ₄	Methane
CO	Carbon monoxide
CO ₂	Carbon dioxide
Pt	Platinum
AC	Activated carbon
Ni	Nickel
Ru	Ruthenium
ZrO ₂	Zirconium oxide
Co	Cobalt
Ti	Titanium
Ir	Iridium
Zi	Zirconium
NaOH	Sodium hydroxide
Na ₂ CO ₃	Sodium carbonate
Ca(OH) ₂	Calcium hydroxide
KOH	Potassium hydroxide
K ₂ CO ₃	Potassium carbonate
H ₂ O ₂	Hydrogen peroxide
γ-Al ₂ O ₃	Gamma alumina
OD	Outer diameter
ID	Inner diameter
CSTR	Continuous stirred tank reactor
HPLC	High-performance liquid chromatography
CMC	Sodium carboxymethyl cellulose

S/V	Surface area-to-volume ratios
LCA	Life cycle assessment
TEA	Technoeconomic analysis

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